Ch 3: Metals and Non-Metals

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Name two metals that are so soft they can be cut with a knife. Name two non-metals that are hard. (PYQ 2019)

Q2. Name one metal and one non-metal that are exceptions to the general physical properties of metals and non-metals. (PYQ 2022)

Q3. Why are metals generally good conductors of heat and electricity? Name one exception to this. (PYQ 2020)

Q4. What is malleability? What is ductility? Which metal is the most malleable? (PYQ 2019)

Q5. What is sonority? Name one metal that is not sonorous and one non-metal that is an exception. (PYQ)

Q6. What is an ore? What is a mineral? How is an ore different from a mineral? (PYQ 2020)

Q7. What is gangue (matrix)? What is the first step in the extraction of metals from their ores? (PYQ 2019)

Q8. What is calcination? Give one example with a balanced equation. (PYQ 2018)

Q9. What is roasting? Give one example with a balanced equation. How is it different from calcination? (PYQ 2019)

Q10. What is thermite reaction? Write the balanced chemical equation. What is its industrial application? (PYQ 2020)

Q11. What is the reactivity series of metals? Name the most reactive and least reactive metals. (PYQ 2019)

Q12. Why does sodium metal have to be stored under kerosene oil? (PYQ 2022)

Q13. Name the gas evolved when: (i) zinc reacts with dilute HCl (ii) sodium reacts with water. Write balanced equations for both. (PYQ 2020)

Q14. What happens when copper is heated in air? Write the balanced chemical equation. What type of oxide is formed? (PYQ 2019)

Q15. What is an amphoteric oxide? Give one example with equations showing its amphoteric nature. (PYQ 2020)

Q16. What is an ionic compound? Give two properties that are characteristic of ionic compounds. (PYQ 2018)

Q17. Why do ionic compounds have high melting and boiling points? (PYQ 2019)

Q18. Why do ionic compounds conduct electricity in the molten state but not in the solid state? (PYQ 2022)

Q19. What is an alloy? Give two examples of alloys and their compositions. (PYQ 2019)

Q20. Why are alloys generally harder and have lower conductivity than pure metals? Give one example. (PYQ 2020)

Q21. What is corrosion? What is the chemical composition of rust? Write the condition required for rusting. (PYQ 2019)

Q22. Name two methods to prevent corrosion of metals. Explain any one method with an example. (PYQ 2020)

Q23. What is galvanisation? How does it prevent corrosion of iron? (PYQ 2022)

Q24. Why do we use aluminum for making electrical wires and cooking utensils? Give two reasons for each use. (PYQ 2019)

Q25. What happens when iron is treated with steam? Write the balanced chemical equation. Name the compound formed. (PYQ 2020)

Q26. Why does gold and silver not react with dilute acids? Name the acid that dissolves gold. What is it called? (PYQ 2021)

Q27. What is the difference between a metal oxide and a non-metal oxide in terms of their acidic/basic nature? Give one example each. (PYQ 2022)

Q28. What are noble metals? Name two noble metals. Why are they not found combined in nature? (PYQ 2019)

Q29. Why is aluminium found as a compound in nature but gold is found in the free state? (PYQ 2020)

Q30. Write the electron-dot structure of sodium chloride (NaCl). What type of bond is present in it? (PYQ 2019)

Q31. What is electrolytic refining? Name the anode, cathode, and electrolyte used in the electrolytic refining of copper. (PYQ 2022)

Q32. Why does the reverse of the following reaction not occur? Zn(s) + CuSO4(aq) -> ZnSO4(aq) + Cu(s). Justify with reason. (PYQ 2023)

SECTION B - 3 MARKS

Q1. Compare the physical properties of metals and non-metals: (PYQ 2019)

Property	Metals	Non-Metals	Exception (if any)
Physical state at room temperature			
Lustre			
Malleability and ductility			
Conductivity (heat and electricity)			
Melting and boiling point			
Density			
Sonority			

Q2. Write the balanced chemical equations for the reactions of metals with the following. State the observation in each case: (PYQ 2020)

Metal	Reacts with	Balanced Equation	Observation
Sodium (Na)	Cold water		
Magnesium (Mg)	Hot water / steam		
Iron (Fe)	Steam		
Copper (Cu)	Dilute HCl		
Zinc (Zn)	Dilute H2SO4		

Q3. Write the balanced chemical equations for the reactions of metals with oxygen. State the nature of oxide formed: (PYQ 2019)

Metal	Balanced Equation with O2	Name of Oxide	Nature of Oxide
Sodium (Na)			
Magnesium (Mg)			
Copper (Cu)			
Iron (Fe)			
Aluminium (Al)			

Q4. Write the balanced chemical equations for the following displacement reactions. State whether the reaction is possible or not and give a reason: (PYQ 2022)

Reaction	Balanced Equation (if possible)	Possible? (Yes/No)	Reason
Zinc + $CuSO_4$ solution			
Copper + $ZnSO_4$ solution			
Iron + $CuSO_4$ solution			
Silver + $FeSO_4$ solution			
Aluminium + $CuSO_4$ solution			

Q5. Write the steps involved in the extraction of metals of different activity levels: (PYQ 2019)

Activity Level of Metal	Examples	Method of Extraction	Key Reaction / Process
Highly reactive (top of reactivity series)			
Moderately reactive (middle of reactivity series)			
Less reactive (bottom of reactivity series)			

Q6. Write the balanced equations and state what happens when the following ores are converted to metals: (PYQ 2020)

Ore / Compound	Process Used	Balanced Equation	Metal Obtained
Cuprous oxide (Cu_2O)	Smelting		
Ferric oxide (Fe_2O_3)	Reduction with CO		
Zinc oxide (ZnO)	Reduction with carbon		
Aluminium oxide (Al_2O_3)	Electrolytic reduction		
Mercuric sulphide (HgS)	Roasting		

Q7. Write the electron-dot structures for the formation of the following ionic compounds. State the type of bond formed: (PYQ 2019)

Ionic Compound	Electron-Dot Structure (description)	Ions Formed	Type of Bond
Magnesium chloride (MgCl2)			
Calcium oxide (CaO)			
Sodium oxide (Na2O)			

Q8. List the properties of ionic compounds. State whether each property applies to solid, molten, or aqueous state: (PYQ 2022)

Property of Ionic Compound	Solid State	Molten State	Aqueous Solution
Electrical conductivity			
Melting and boiling point			
Solubility in water			
Solubility in organic solvents			
Physical state			

Q9. Write the differences between calcination and roasting. Give one example of each with a balanced chemical equation: (PYQ 2018)

Basis of Difference	Calcination	Roasting
Definition		
Type of ore used		
Gas produced		
Example ore		
Balanced equation		

Q10. What is meant by enrichment (concentration) of ore? Write three methods used to concentrate ores and for which type of ore each method is used: (PYQ 2020)

Method of Concentration	Principle Used	Suitable for Which Type of Ore
Hydraulic washing (gravity separation)		

Method of Concentration	Principle Used	Suitable for Which Type of Ore
Froth flotation		
Magnetic separation		

Q11. Write the balanced equations for the following reactions. State what type of chemical change is occurring: (PYQ 2019)

Reaction	Balanced Chemical Equation	Type of Change
Iron with dilute sulphuric acid		
Copper with silver nitrate solution		
Zinc oxide with sodium hydroxide		
Aluminium with iron oxide (thermite)		
Magnesium with nitrogen		

Q12. What is corrosion? Explain the conditions necessary for rusting of iron. Write two methods to prevent rusting with explanation: (PYQ 2022)

Q13. Write the balanced chemical equations for the following reactions. Also identify the oxidising agent in each: (PYQ 2020)

Reaction	Balanced Equation	Oxidising Agent
Copper heated in air		
Zinc heated in air		
Iron heated in steam		
Magnesium heated in nitrogen gas		
Aluminium heated in oxygen		

Q14. Name the following and write their chemical formulae: (PYQ 2022)

Description	Name of Compound	Chemical Formula
Ore of iron from which iron is extracted		

Description	Name of Compound	Chemical Formula
Ore of aluminium		
Ore of copper		
Ore of zinc		
Compound formed when iron rusts		
Compound formed when copper corrodes (green coating)		

Q15. Write the chemical equations and state the observation for the following reactions: (PYQ 2023)

Reaction	Balanced Equation	Observation
Aluminium powder + iron oxide (thermite reaction)		
Iron nail placed in copper sulphate solution		
Copper powder heated in china dish with oxygen		
Zinc added to dilute hydrochloric acid		
Sodium metal dropped in cold water		

Q16. What are alloys? Write the composition, properties, and uses of the following alloys: (PYQ 2019)

Alloy	Composition	Important Properties	Uses
Brass			
Bronze			
Solder			
Steel (stainless)			
Amalgam			

Q17. Write the differences between metals and non-metals based on their chemical properties: (PYQ 2018)

Chemical Property	Metals	Non-Metals
Reaction with oxygen		
Nature of oxide		
Reaction with water		
Reaction with dilute acids		
Reaction with chlorine		
Reaction with hydrogen		

Q18. What is the activity series of metals? List five metals in decreasing order of reactivity. State what happens when each metal reacts with cold water, steam, and dilute HCl: (PYQ 2020)

Metal (in decreasing reactivity)	Reacts with Cold Water?	Reacts with Steam?	Reacts with Dilute HCl?
Potassium (K)			
Sodium (Na)			
Magnesium (Mg)			
Zinc (Zn)			
Copper (Cu)			

Q19. What is electrolytic refining? Draw a labelled diagram and explain the process for copper. Write the reactions at the anode and cathode: (PYQ 2022)

Q20. Write three differences between ionic compounds and covalent compounds: (PYQ 2019)

Basis of Difference	Ionic Compounds	Covalent Compounds
Type of bond		
Melting and boiling point		
Electrical conductivity		
Solubility in water		
State at room temperature		

Q21. What is the role of the following in metallurgy? (PYQ 2020)

Process / Agent	Role in Metallurgy	Example with Equation
Coke (carbon) as reducing agent		
Flux		
Slag		

Q22. Write the sequence of steps involved in extracting a metal from its sulphide ore (moderately reactive metal). Give equations for each step using copper as an example: (PYQ 2019)

Step No.	Step Name	Process Description	Balanced Equation
Step 1	Concentration of ore		
Step 2	Roasting		
Step 3	Reduction / Smelting		
Step 4	Refining (electrolysis)		

Q23. Explain why: (PYQ 2022)

Statement	Explanation
Sodium is stored in kerosene	
Aluminium forms a protective oxide layer	
Gold and platinum are used to make jewellery	
Copper vessels develop a green coating over time	
Iron articles are coated with zinc (galvanised)	

Q24. Name the following and give one use for each: (PYQ 2023)

Description	Name	One Use
An ore of iron used in blast furnace		
A liquid metal at room temperature		
The hardest natural substance (non-metal allotrope)		

Description	Name	One Use
A good conductor of electricity that is a non-metal		
The most abundant metal in earth's crust		

Q25. Write balanced equations for the reactions of non-metals with the following substances: (PYQ 2019)

Non-metal	Reacts with	Balanced Equation	Nature of Product
Carbon (C)	Oxygen		
Sulphur (S)	Oxygen		
Nitrogen (N2)	Oxygen (at high temp)		
Hydrogen (H2)	Chlorine		
Carbon (C)	Hydrogen		

Q26. What happens when: (i) dilute HCl is added to a mixture of iron and sulphur (ii) iron and sulphur are heated together and then dilute HCl is added. State observations and write equations. (PYQ 2020)

Q27. What is the difference between a mineral and an ore? Give two examples of ores of iron and state the name of the ore used commercially. (PYQ 2019)

Q28. Write balanced equations for the following. Also state the type of reaction in each: (PYQ 2024)

Reaction	Balanced Equation	Type of Reaction
Magnesium + oxygen		
Copper + oxygen (on heating)		
Iron + steam		
Zinc oxide + carbon		
Copper oxide + hydrogen		

Q29. How is aluminium extracted from bauxite? Write the chemical equations for: (i) purification of bauxite using NaOH (ii) electrolytic decomposition of alumina.

(PYQ 2019)

Q30. State with reasons: (i) Why is copper used for making electrical wires? (ii) Why is aluminium used for making aircraft bodies? (iii) Why is tungsten used in electric bulb filaments? (PYQ 2020)

SECTION C - 5 MARKS

Q1. Write the balanced chemical equations for the reactions of the following metals with dilute hydrochloric acid. State the observation and name the gas evolved in each case: (PYQ 2019)

Metal	Balanced Equation with Dilute HCl	Observation	Gas Evolved
Magnesium (Mg)			
Aluminium (Al)			
Iron (Fe)			
Zinc (Zn)			
Copper (Cu)			

Q2. Describe the extraction of metals at different activity levels. Write the method of extraction, key processes, and balanced equations for each: (PYQ 2022)

Activity Level	Examples of Metals	Method of Extraction	Key Process and Balanced Equation
Very high reactivity (K, Na, Ca, Mg, Al)			
Moderate reactivity (Zn, Fe, Pb, Cu)			
Low reactivity (Hg, Ag, Au, Pt)			

Q3. Write the differences between metals and non-metals based on both physical and chemical properties. Give two examples and one balanced equation for each property: (PYQ 2020)

Property	Metals	Non-Metals	Example / Equation
Lustre			
Malleability			
Conductivity			
Reaction with oxygen (oxide nature)			
Reaction with water			

Property	Metals	Non-Metals	Example / Equation
Reaction with dilute acid			
Reaction with hydrogen			

Q4. Describe the steps involved in the extraction of iron from haematite (Fe_2O_3) in a blast furnace. Write balanced equations for all reactions occurring at different zones: (PYQ 2019)

Zone in Blast Furnace	Temperature	Reaction Occurring	Balanced Equation
Combustion zone (bottom)			
Reduction zone 1 (upper)			
Reduction zone 2 (middle)			
Limestone decomposition zone			
Slag formation zone			

Q5. Write balanced equations for the following reactions. Identify the oxidising agent and reducing agent in each: (PYQ 2020)

Reaction	Balanced Equation	Oxidising Agent	Reducing Agent
Zinc oxide + carbon -> zinc + carbon monoxide			
Copper oxide + hydrogen -> copper + water			
Iron oxide + aluminium -> iron + aluminium oxide			
Lead oxide + carbon -> lead + carbon dioxide			
Ferric oxide + carbon monoxide -> iron + CO_2			

Q6. What is the reactivity series? Arrange the following metals in decreasing order of reactivity. Predict whether a reaction will occur between each pair and write the balanced equation if it does: (PYQ 2022)

Metal A	Metal Salt Solution B	Will Reaction Occur? (Yes/No)	Reason	Balanced Equation (if yes)
Zinc	$CuSO_4$ solution			
Copper	$ZnSO_4$ solution			
Iron	$CuSO_4$ solution			
Copper	$AgNO_3$ solution			
Silver	$CuSO_4$ solution			

Q7. A student performs the following experiments. State the observation, write balanced equations, and name the type of reaction: (PYQ 2023)

Experiment	Observation	Balanced Equation	Type of Reaction
Iron nail placed in blue $CuSO_4$ solution			
Sodium dropped in cold water			
Magnesium ribbon burnt in oxygen			
Thermite reaction: Al + Fe_2O_3 (heated)			
Zinc added to NaOH solution			

Q8. Explain corrosion. Write the conditions required for rusting of iron. Write five methods to prevent corrosion with a brief explanation of each: (PYQ 2019)

Method of Preventing Corrosion	How It Works	Example
Painting or varnishing		
Galvanisation		
Alloying		
Electroplating		
Oil and grease application		

Q9. Complete the following table about the occurrence and extraction of important metals: (PYQ 2022)

Metal	Most Common Ore	Formula of Ore	Method of Extraction	One Important Use
Iron (Fe)				
Aluminium (Al)				
Copper (Cu)				
Zinc (Zn)				
Sodium (Na)				

Q10. Write the balanced equations for the reactions of metals with water (cold water, hot water, and steam). State what is produced and how vigorously the reaction occurs: (PYQ 2020)

Metal	Condition (Cold / Hot / Steam)	Balanced Equation	Products	Vigour of Reaction
Potassium (K)	Cold water			
Sodium (Na)	Cold water			
Calcium (Ca)	Cold water			
Magnesium (Mg)	Hot water / steam			
Iron (Fe)	Steam only			

Q11. Write the electron-dot structures for the formation of the following ionic compounds. State the charge on each ion formed: (PYQ 2019)

Compound	Metal Ion (symbol + charge)	Non-metal Ion (symbol + charge)	Electron-Dot Structure (description)	Formula
Sodium chloride				
Magnesium oxide				
Calcium chloride				
Aluminium oxide				
Sodium oxide				

Q12. Write the differences between the following pairs: (PYQ 2022)

Basis	Mineral	Ore
Definition		
Contains metal profitably?		
Example		

Basis	Calcination	Roasting
Definition		
Type of ore used		
Gas produced		
Balanced equation		

Q13. Answer the following questions about metals and non-metals: (PYQ 2023)

Question	Answer
Name two metals that float on water. Why?	
Why is aluminium extracted by electrolysis not by reduction with carbon?	
Why do sodium and potassium catch fire if kept in open air?	
Name the metal that is stored in water. Why?	
Why is diamond used as a cutting tool while graphite is used as a lubricant?	

Q14. Write the balanced equations and give observations for the following. Also state what type of oxide is formed: (PYQ 2024)

Reaction	Balanced Equation	Observation	Type of Oxide	Nature (Acidic/Basic/Amphoteric)
Sulphur burns in oxygen				
Nitrogen reacts with oxygen				
Sodium reacts with oxygen				
Aluminium reacts with oxygen				

Reaction	Balanced Equation	Observation	Type of Oxide	Nature (Acidic/Basic/Amphoteric)
Carbon burns in excess oxygen				

Q15. Case Study: A student observes different metals reacting with dilute HCl, water, and oxygen. Complete the following table and answer the questions below: (PYQ 2024 case-based)

Metal	Reaction with Dilute HCl	Reaction with Water / Steam	Reaction with Oxygen	Position in Reactivity Series
Potassium (K)				
Magnesium (Mg)				
Zinc (Zn)				
Copper (Cu)				
Gold (Au)				